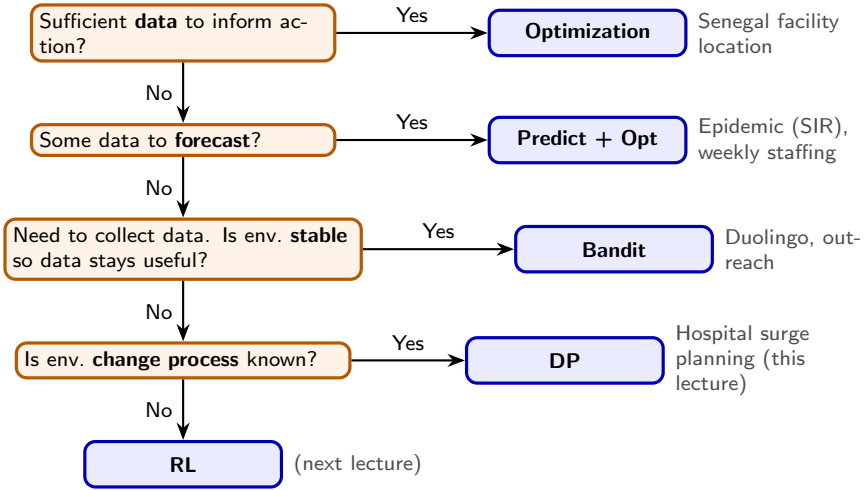
Lecture 4: Dynamic Programming and Backward Induction

Balancing Exploration (Data) and Exploitation (Action) When Things Change

Peter Zhang

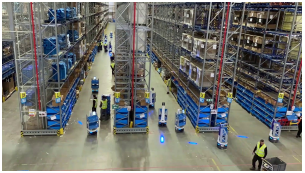
12768/19867/94867: From Data to Action

Recap of Course So Far: How to Go From Data to Actions



1. Workforce planning: base and surge
2. Optional: the same method, valuing an acquisition

Workforce Planning: A Warehouse in the Netherlands



GXO warehouse operations. Photo: Locus Robotics, via PR Newswire.

GXO runs H&M's warehouse in the Netherlands. Part of the floor is now robot-run, volume is severely seasonal. Multiple steps: unboxing, stocking, picking, packing, shipping.

Capacity gets committed at several horizons:

Commit	**What you fix**	**$/hr**	**Data for Decision**
quarterly	permanent FTE	lowest	the season
∼4 weeks	temp intake	low	month forecast
day-of	overtime, flex	**highest**	**actual orders**

Cheap capacity has to be committed *before* you know. Expensive capacity can wait *until* you do.

Workforce Planning: A Hospital in Pennsylvania



Reading Hospital, West Reading, PA.

Doctors are staffed on long-term contracts, a fixed number per hour. Potentially long waits at some hours of the day and idle doctors at others. Patients go through: triage, beds, imaging, discharge.

Capacity gets committed at several horizons:

Commit	**What you fix**	**$/hr**	**Data for Decision**
annual	physician contracts	lowest	the season
∼6 weeks	published nurse roster	low	month census
∼1 week	swaps, extra shifts	medium	week ahead
day-of	per-diem, float, overtime	**highest**	**the waiting room**

Seasonality does not necessarily cause day-of shifts. If you had perfect prediction, you can always preplan with more effective and inexpensive options.

The Simple Model

1. Pick base staff weeks ahead (e.g., based on SIR prediction; cheap)
2. Update with short-term forecast day-of
3. Add surge staff (expensive, flexible)

Goal: balance staffing cost vs. patient waiting.

Potential Impact of Staffing

- Reading Hospital in West Reading, PA, reportedly saved $1M per year after implementing new staffing policies in 2021
- **Cost Effectiveness:** Carri Chan and coauthors show that a prediction-driven *base + surge* staffing approach at NYP can reduce annual costs by 10–16% ($2–3M) while maintaining timely access
- **Patient experience:** that well-timed surge staffing can cut waiting times by up to ˜20–30 minutes for each patient

Yue Hu, Carri W. Chan , Jing Dong (2024) Prediction-Driven Surge Planning with Application to Emergency Department Nurse Staffing. Management Science 71(3):2079-2126.

Diving into The Details

Waiting in the Hospital

- From conversations with the hospital's leadership, major opportunities seem to exist on *supply-side management*
- The mismatch is *within* the day, not just across the season: high wait times at some hours and idle doctors at others
- The new lever is on-demand staffing: additional doctors can be added *in a given hour*, at a price premium
- So the question is not "how many doctors" but **how many, in which hour, in which state of the waiting room**

The Technical Solution

We need a "Policy", a complete contingency plan that tells us what to do at each time period, under any possible state:

Number of on-demand doctors to staff in each hour as a function of number of patients waiting

Time

System State	$t=0$	$t=1$	$t=2$	$t=3$	...
S_0	?	?	?	?	
S_1	?	?	?	?	
S_2	?	?	?	?	
S_3	?	?	?	?	
$\vdots$					

Comparing with Bandit Setting

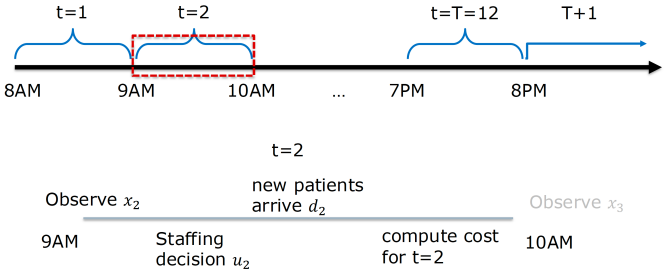
- In the basic multi-armed bandit settings, we assumed that data collected about each arm remain useful for a long time. E.g., the quality of each option does not change over time.
- Duolingo's case hinted that in practice, this assumption may be restrictive – the same notification push may become less effective if user is exposed to it many times.
- Here, the same "environment change" happens: the effect of having 5 doctors is case by case.
  - 5 doctors when there is no patient waiting...v.s.
  - 5 doctors when there are 50 patients waiting...

Problem Setup

- **Hospital Operations**
  - The hospital operates for 12 hours between 8am and 8pm
  - A day starts with an empty waiting room
- **Doctor Supply**
  - Each doctor can treat exactly 2 patients per hour
  - A fixed staff of 10 doctors for the full 12-hour period
  - Each additional doctor can be staffed hourly at a cost of p
- **Patient Demand**
  - Patients arrive at each hour (may be random)
- **Waiting Costs**
  - At the beginning of each hour, you incur cost w per waiting patient
  - At the end of the day, you incur cost W for any remaining patient

Sequence of Events and Cost Accumulation

- **Stages** $t = 1, \ldots, T$: hours of a day
- **States** x_t: number of patients in the waiting room in hour t
- **Actions** u_t: number of additional doctors staffed in hour t
- **Parameters** d_t: patient demand in hour t
- **What Happens per Stage?**



Sequence of Events and Cost Accumulation

- **Stages** $t = 1, \ldots, T$: hours. **States** x_t: patients waiting. **Actions** u_t: extra doctors. **Parameters** d_t: demand.
- **Transition Function**:

$$x_{t+1} = f_t(x_t, u_t) = \max\left[0,\ x_t + d_t - 2(10 + u_t)\right]$$

  characterization of state in next period, given the state of the system and the decision made in the current period
- **Cost Function**:

$$g_t(x_t, u_t) = w \cdot x_t + p \cdot u_t \qquad \text{cost per stage}$$
$$g_{T+1}(x_{T+1}) = W \cdot x_{T+1} \qquad \text{termnial cost}$$

- **Bellman Equation**: Cost of state x_t in hour t

$$J_t(x_t) = \min_{u_t \geq 0} \left[g_t(x_t, u_t) + J_{t+1}(x_{t+1}) \right]$$
$$J_{T+1}(x_{T+1}) = W\ x_{T+1}$$

Toy Numerical Example: Terminal Cost at $T+1$

Definition:

$$J_{T+1}(x_{T+1}) = W \cdot x_{T+1}, \quad W = \$200 \text{ per patient}$$

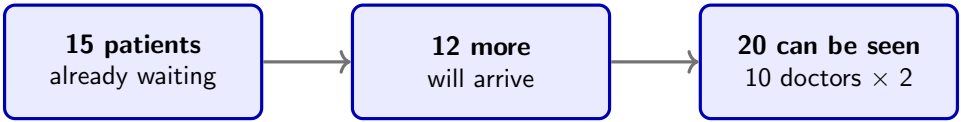
Terminal cost if x_{T+1} patients remain at end of day:

x_{T+1} (patients waiting)	$J_{T+1}(x_{T+1}) = 200 \times x_{T+1}$
0	\$0
1	\$200
2	\$400
3	\$600
4	\$800
5	\$1,000
6	\$1,200
7	\$1,400
⋮	⋮

The Last Hour: What We Know

Suppose it is 7pm, the last hour, and **15** patients are waiting.

15 is just one case. We will redo this for every other queue length.



$15 + 12 - 20 =$ **7** patients left at 8pm, *if we do nothing.*

Each extra doctor costs **$250** and clears **2** more patients.
Each patient still waiting at 8pm costs **$200**.

How many surge doctors?

The Last Hour: Three Costs, Only Two of Them Move

Waiting now

$w\,x_T = \$30 \times 15$

\$450

already spent, whatever we do

Surge we buy

$p\,u = \$250 \times u$

\$250 each

one more doctor: **pay \$250**

Left at 8pm

$W\,(7 - 2u)^+$

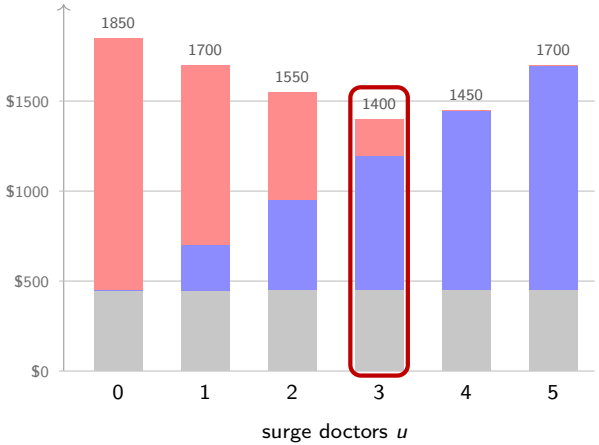
\$200 each

one more doctor clears 2: **save \$400**

$$\text{total}(u) = \underbrace{450}_{\text{waiting, unavoidable}} + \underbrace{250\,u}_{\text{what surge costs}} + \underbrace{200\,(7 - 2u)^+}_{\text{penalty at 8pm}}$$

Buying a doctor costs \$250 and saves \$400 of penalty, **until the waiting room is empty**. After that it saves nothing.

The Last Hour: Try Every u



u	wait	surge	left	**total**
0	450	0	1400	1850
1	450	250	1000	1700
2	450	500	600	1550
3	**450**	**750**	**200**	**1400**
4	450	1000	0	1450
5	450	1250	0	1700

waiting surge left at 8pm

$$u_T^\star = 3 \qquad \text{and so} \qquad J_T(15) = \$1{,}400$$

That one number summarizes the entire rest of the day, from this state.

Toy Numerical Example: Two Tables to Fill In

Best action $u_t^\star(x)$

x	t=1	t=2	$\cdots$	t=11	t=12
0	?	?		?	?
1	?	?		?	?
2	?	?		?	?
$\vdots$					
15	?	?		?	**3**
16	?	?		?	?

Cost-to-go $J_t(x)$

x	t=1	$\cdots$	t=11	t=12	t=13
0	?		?	?	0
1	?		?	?	200
2	?		?	?	400
$\vdots$					$\vdots$
15	?		?	**1400**	3000
16	?		?	?	3200

- The **green** column is $t = T + 1$. We never choose anything there, we just read off $J_{T+1}(x) = Wx$. That is the only column we get for free.
- The **red** cells are the one calculation we just did. Two numbers out of a table with hundreds of cells.
- Everything else is a question mark. We fill it **right to left**, one column at a time.

Finishing the Last Column: Every Other x

Same three costs, same six choices of u, only the starting queue changes:

$$\text{total}(u) = \underbrace{30\,x}_{\text{waiting}} + \underbrace{250\,u}_{\text{surge}} + \underbrace{200\,(x-8-2u)^+}_{\text{left at 8pm}}$$

x	left over if u=0	$u_T^\star(x)$	$J_T(x)$
0	0	0	\$0
⋮			
8	0	0	\$240
9	1	0	\$470
10	2	1	\$550
11	3	1	\$780
12	4	2	\$860
13	5	2	\$1,090
14	6	3	\$1,170
15	**7**	**3**	**\$1,400**
16	8	4	\$1,480

The shape of the answer:

- Up to $x = 8$, the last hour already has enough capacity. Buy nothing.
- Past that, buy roughly **one doctor for every 2 patients over the line**: \$250 spent to save \$400.
- Except at $x = 9$: only **one** patient is left over, so a doctor saves \$200, not \$400. Not worth \$250.

None of this was assumed. It *fell out* of trying every u.

That is the whole $t = 12$ column. Now $t = 11$ can use it.

General Computations

- Initialize Cost-to-Go Function with **Terminal Cost** (end of day cost)

$$J_{T+1}(x_{T+1}) = 200\, x_{T+1} = 200 \times (\#\text{ of remaining patients})$$

- Calculate **Immediate** and **Future Costs** for each decision

$$g_t(x_t, u_t) = \text{Waiting Cost} + \text{Staffing Cost} = 30\, x_t + 250\, u_t$$

- Choose the optimal action u_t^* to balance immediate and future costs (cost-to-go)

$$\begin{aligned} u_t^* &= \min_{u_t \geq 0} \Big[g_t(x_t, u_t) + J_{t+1}(x_{t+1}) \Big] \\ &= \min_{u_t \geq 0} \Big[30\, x_t + 250\, u_t + J_{t+1}\big(\max\big[0,\, x_t + d_t - 2(10 + u_t)\big]\big) \Big] \end{aligned}$$

LIVE DEMO

Watch the table fill in, from the last hour backwards

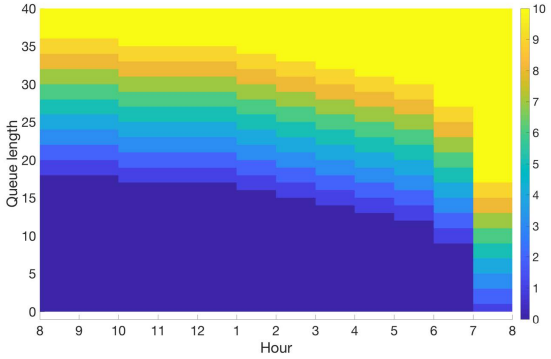
`bi-bench.html` · tab 1, Hospital

Solve backward. Click a cell, then step through the candidate u one at a time.

Then change the day: redraw the arrivals, or switch them from deterministic to random. *Pin* a policy and the table shows you what moved.

"Dynamic" Results

- Hire additional doctors if...
  - > 15 patients waiting in early stages of the day
  - Waiting room is non-empty towards the end of the day



Color shows the optimal decision (number of surge doctors to staff).

1. Workforce planning: base and surge
2. Optional: the same method, valuing an acquisition

Samsung's Acquisition of Harman in 2017

Optional. Nothing new is required here. It is a second look at the same recursion, in a setting where nobody makes a decision in the middle, so the min disappears and only the expectation is left.

Samsung Electronics sought entry into the connected car and automotive electronics market, while Harman International Industries specialized in infotainment, telematics, and automotive software,
which would potentially deliver **dynamic capabilities-based synergies**.

SAMSUNG HARMAN
A SAMSUNG COMPANY

Čirjevskis, A. (2021). Exploring the Link of Real Options Theory with Dynamic Capabilities Framework in Open Innovation-Type Merger and Acquisition Deals [Publisher: Multidisciplinary Digital Publishing Institute]. *Journal of Risk and Financial Management*, *14*(4), 168. https://doi.org/10.3390/jrfm14040168

The Business Question

How to value the dynamic capabilities-based synergies?

- Treat acquisition as a **call option**[1] on the future market value
  - Call option is a contract between the buyer and the seller of the call option to exchange a security at a fixed price
  - Final payoff is upside if market value exceeds strike, otherwise zero

$$C_T(S_T) = \max(S_T - K, 0)$$

  - S_T: underlying assets value of Samsung + Harman when call option is mature
  - K: combined stand-alone value without the deal

[1] https://en.wikipedia.org/wiki/Call_option

The Intuition

By taking the **expected future payoffs** of the acquisition option, synergies can be valued as the **present option value**, i.e. synergies

- Consider synergies as an investment, the present option value represents today's stake, which yields uncertain future returns, discounted at an interest rate
- If you invest \$100 with a compound interest rate of 15% for 2 years, the payoff is $100 * 1.15^2 = 132.25$
- Conversely, knowing a payoff of \$132.25 and interest rate, we can **back out** the initial investment $132.25/1.15^2 = 100$

In this case, "**back out**" implements **Backward Induction**

Forward: Possible Option Prices with Binomial Tree

- S_t is price of the underlying assets at time step t
  - $t = 0$ is the present
  - $t = T$ is call option maturity
- At each time step, the underlying value S_t is either
  - increased by an up factor u with a risk-neutral probability p
  - decreased by a down factor d with a probability $(1 - p)$

$$S_{t+1} = \begin{cases} uS_t & \text{with probability } p \textit{ (value up)} \\ dS_t & \text{with probability } 1 - p \textit{ (value down)} \end{cases}$$

- At time step t, underlying value with n ups in total is

$$S_{t,n} = S_0 \; u^n \; d^{t-n}$$

Forward: Possible Option Prices with Binomial Tree

$$S_{t+1} = \begin{cases} uS_t & \text{with probability } p \textit{ (value up)} \\ dS_t & \text{with probability } 1-p \textit{ (value down)} \end{cases}$$

S_0	S_1	S_2	S_3	S_4	S_5
					600.73
				480.47	
			384.28		384.28
		307.34 (up twice)		307.34	
	245.81 (up)		245.81		245.81
196.60		196.60 (up and down)		196.60	
	157.24 (down)		157.24		157.24
		125.76 (down twice)		125.76	
			100.58		100.58
				80.45	
					64.34

Backward Induction: Real Option Valuation

- **Terminal Condition**: At maturity (T), payoff is

$$C_T(S_T) = \max(S_T - K, 0)$$

- **Bellman Equation**

$$C_t(S_t) = e^{-r\Delta t}\left(p\, C_{t+1}(uS_t) + (1-p)\, C_{t+1}(dS_t)\right)$$

  - $t = 0, 1, \ldots, T-1$
  - r is the risk-free interest rate
- **Synergy Value**: $C_0(S_0)$

LIVE DEMO

The same recursion, on a tree

`bi-bench.html` · tab 2, Samsung

Build forward, then roll back. Click a node for its arithmetic.

Backward Induction: Real Option Valuation

$C_T(S_T) = \max\Big(S_T - K, 0\Big)$, where K is a constant $C_t(S_t) = e^{-r\Delta t}\Big(p\, C_{t+1}(uS_t) + (1-p)\, C_{t+1}(dS_t)\Big)$

$C_0(S_0)$	$C_1(S_1)$	$C_2(S_2)$	$C_3(S_3)$	$C_4(S_4)$	$C_5(S_5)$	S_5
					380.33	600.73
				262.91		
			169.52		163.88	384.28
		104.20		89.78		
	61.89		48.12		25.42	245.81
35.78		25.37		11.88		
	13.20		5.55		0	157.24
		2.59		0		
			0		0	100.58
				0		
					0	64.34

Key Insights

- The result of $35.78M synergies shows that acquiring Harman is in the money
- Backward Induction allows Samsung to **roll back future uncertainty** into today's decision
- This case is a forward-backward application without intermediate decision-making